

## DAYTON WRIGHT BROTHERS, OH, USA

WMO: 724276

Lat: 39.594N

Lon: 84.226W

Elev: 293

StdP: 97.85

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 53859

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | -15.4      | -12.5 | -20.2                         | 0.6 | -14.5 | -17.6 | 0.8 | -11.6 | 10.8                  | 3.1  | 9.4 | 1.9  | 2.8                   | 280  | 0.477 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 9.8                    | 32.4            | 23.1 | 31.2 | 22.6 | 29.9 | 22.0 | 24.7                | 30.0 | 24.0 | 28.8 | 23.3 | 27.8 | 3.8                  | 230  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 22.9                            | 18.3 | 27.2 | 22.4 | 17.8 | 26.7 | 22.0 | 17.2 | 26.2 | 76.2          | 30.0 | 73.2 | 29.1 | 70.6 | 27.7 | 28.6              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| (4) 9.0           | 8.1  | 7.3 | -18.5                      | 34.0 | 3.6                | 1.7 | -21.1                                              | 35.2 | -23.2      | 36.2 | -25.2      | 37.1 | -27.8      | 38.4 |    |    |
| (5)               |      |     | -19.2                      | 25.8 | 3.4                | 0.9 | -21.6                                              | 26.5 | -23.6      | 27.0 | -25.5      | 27.5 | -28.0      | 28.2 |    |    |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 12.0   | -1.6  | 0.3   | 5.7   | 12.2  | 17.7  | 22.0  | 23.7  | 23.1  | 19.6  | 13.0  | 6.7   | 1.2  |
|                                                                                  | DBStd       | 10.24  | 6.71  | 6.41  | 6.14  | 5.13  | 4.58  | 3.14  | 2.86  | 2.70  | 4.02  | 5.05  | 5.34  | 5.85 |
|                                                                                  | HDD10.0     | 1272   | 362   | 278   | 164   | 36    | 3     | 0     | 0     | 0     | 0     | 26    | 127   | 277  |
|                                                                                  | HDD18.3     | 2878   | 617   | 506   | 394   | 192   | 70    | 7     | 1     | 2     | 33    | 178   | 349   | 531  |
|                                                                                  | CDD10.0     | 2016   | 3     | 6     | 30    | 102   | 241   | 361   | 426   | 407   | 288   | 119   | 28    | 5    |
|                                                                                  | CDD18.3     | 578    | 0     | 0     | 1     | 7     | 49    | 118   | 168   | 150   | 70    | 13    | 1     | 0    |
|                                                                                  | CDH23.3     | 4721   | 0     | 1     | 8     | 59    | 388   | 925   | 1460  | 1205  | 578   | 95    | 1     | 0    |
|                                                                                  | CDH26.7     | 1472   | 0     | 0     | 1     | 4     | 88    | 280   | 508   | 385   | 186   | 21    | 0     | 0    |
| Wind                                                                             | WSAvg       | 3.4    | 4.1   | 4.0   | 4.0   | 4.1   | 3.3   | 3.0   | 2.7   | 2.6   | 2.8   | 3.3   | 3.6   | 3.8  |
| Precipitation                                                                    | PrecAvg     | 1053   | 79    | 58    | 87    | 100   | 109   | 111   | 102   | 82    | 87    | 83    | 67    | 86   |
|                                                                                  | PrecMax     | 1520   | 204   | 120   | 195   | 262   | 178   | 199   | 160   | 163   | 207   | 140   | 157   | 149  |
|                                                                                  | PrecMin     | 751    | 32    | 29    | 31    | 42    | 37    | 47    | 66    | 46    | 24    | 35    | 17    | 41   |
|                                                                                  | PrecStd     | 160    | 44    | 30    | 35    | 48    | 45    | 40    | 28    | 31    | 51    | 35    | 34    | 27   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 17.1  | 19.9  | 24.4  | 27.3  | 30.9  | 32.8  | 34.4  | 33.5  | 33.1  | 29.2  | 22.2  | 17.4 |
|                                                                                  |             | MCWB   | 14.2  | 14.9  | 16.0  | 17.9  | 21.9  | 22.8  | 24.7  | 23.1  | 21.9  | 20.8  | 15.6  | 14.3 |
|                                                                                  | 2%          | DB     | 14.1  | 15.8  | 21.0  | 25.1  | 28.9  | 31.1  | 32.5  | 31.9  | 31.0  | 25.8  | 19.1  | 14.6 |
|                                                                                  |             | MCWB   | 11.9  | 12.1  | 14.4  | 16.3  | 20.4  | 22.3  | 24.1  | 22.8  | 21.3  | 18.7  | 14.0  | 12.2 |
|                                                                                  | 5%          | DB     | 11.7  | 12.9  | 17.9  | 22.9  | 27.5  | 29.7  | 31.2  | 30.3  | 28.8  | 23.6  | 17.4  | 12.4 |
|                                                                                  |             | MCWB   | 9.6   | 9.9   | 12.6  | 15.2  | 19.5  | 21.6  | 23.2  | 22.2  | 20.8  | 17.4  | 13.3  | 10.2 |
|                                                                                  | 10%         | DB     | 8.5   | 10.1  | 15.7  | 21.0  | 25.7  | 28.2  | 29.7  | 28.8  | 27.0  | 21.4  | 15.1  | 9.9  |
|                                                                                  |             | MCWB   | 5.9   | 6.9   | 11.4  | 14.5  | 18.9  | 20.9  | 22.6  | 21.7  | 19.7  | 16.4  | 11.5  | 7.3  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 14.8  | 15.5  | 17.4  | 19.7  | 22.9  | 24.6  | 26.0  | 25.2  | 24.0  | 22.0  | 17.3  | 15.2 |
|                                                                                  |             | MCDB   | 16.6  | 19.3  | 22.7  | 25.4  | 29.0  | 29.8  | 32.0  | 30.7  | 29.3  | 26.8  | 19.9  | 16.8 |
|                                                                                  | 2%          | WB     | 12.6  | 13.0  | 15.4  | 18.0  | 21.7  | 23.6  | 25.1  | 24.3  | 22.9  | 20.4  | 15.5  | 12.7 |
|                                                                                  |             | MCDB   | 13.9  | 15.0  | 19.3  | 22.1  | 27.1  | 28.8  | 30.8  | 29.3  | 28.1  | 24.4  | 18.3  | 14.1 |
|                                                                                  | 5%          | WB     | 9.9   | 10.3  | 13.9  | 16.7  | 20.7  | 22.9  | 24.3  | 23.6  | 21.9  | 18.7  | 13.9  | 10.5 |
|                                                                                  |             | MCDB   | 11.3  | 13.0  | 17.3  | 20.6  | 25.6  | 27.9  | 29.3  | 28.1  | 27.0  | 22.3  | 16.6  | 12.2 |
|                                                                                  | 10%         | WB     | 6.1   | 6.8   | 11.9  | 15.5  | 19.7  | 22.1  | 23.6  | 22.9  | 21.0  | 17.0  | 11.9  | 7.7  |
|                                                                                  |             | MCDB   | 8.4   | 9.7   | 15.2  | 19.6  | 24.0  | 26.5  | 28.0  | 26.9  | 24.9  | 20.6  | 14.7  | 9.5  |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 7.9   | 8.6   | 9.5   | 11.0  | 10.4  | 10.0  | 9.8   | 10.1  | 10.8  | 10.2  | 9.1   | 7.2  |
|                                                                                  |             | MCDBR  | 10.0  | 11.9  | 12.4  | 13.3  | 11.9  | 11.5  | 10.7  | 11.4  | 12.7  | 12.3  | 11.4  | 9.7  |
|                                                                                  |             | MCWBR  | 8.9   | 9.6   | 7.8   | 7.6   | 5.5   | 5.1   | 4.2   | 4.5   | 5.3   | 6.6   | 7.8   | 8.6  |
|                                                                                  | 5% WB       | MCDBR  | 9.6   | 11.4  | 10.9  | 11.0  | 10.2  | 9.8   | 9.7   | 9.3   | 10.2  | 10.3  | 9.6   | 9.3  |
|                                                                                  |             | MCWBR  | 9.2   | 9.9   | 7.5   | 7.3   | 5.2   | 5.0   | 4.2   | 4.3   | 4.8   | 6.0   | 7.7   | 9.0  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.300  | 0.323 | 0.346 | 0.389 | 0.434 | 0.444 | 0.447 | 0.442 | 0.392 | 0.352 | 0.322 | 0.300 |      |
|                                                                                  | taud        | 2.462  | 2.364 | 2.378 | 2.268 | 2.184 | 2.202 | 2.202 | 2.198 | 2.315 | 2.428 | 2.472 | 2.476 |      |
|                                                                                  | Ebn at Noon | 873    | 894   | 909   | 885   | 849   | 838   | 831   | 825   | 848   | 851   | 837   | 844   |      |
|                                                                                  | Edh at Noon | 79     | 100   | 110   | 131   | 146   | 143   | 142   | 139   | 116   | 93    | 78    | 73    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 1.73   | 2.57  | 3.58  | 4.84  | 5.49  | 6.08  | 5.91  | 5.46  | 4.48  | 3.18  | 2.15  | 1.47  |      |
|                                                                                  | RadStd      | 0.15   | 0.30  | 0.31  | 0.39  | 0.46  | 0.39  | 0.35  | 0.28  | 0.42  | 0.30  | 0.25  | 0.15  |      |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) | Regional (35 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | +118    | +70     |

Nomenclature: See separate page